The University of Nottingham
Malaysia Campus

SCHOOL OF COMPUTER SCIENCE

A LEVEL 1 MODULE, AUTUMN SEMESTER 2015-2016

MATHEMATICS FOR COMPUTER SCIENTISTS

Time allowed ONE Hour

Candidates must NOT start writing their answers until told to do so

There are THREE questions in the exam paper. Answer all THREE questions.

Only silent, self-contained calculators with a single-line display are permitted in this examination.

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

K: knowledge
C: Comprehension
P: Problem Solving

1. Questions on Logic, Sets, Functions, Matrices, and Relations.

(a) Let A be the set $\{1,2,3\}$. The following relations are subsets of $A \times A$,

$R_1 = \{(x,y) | \, x \in A \wedge y \in A \wedge (x - y = 1)\}$,

$R_2 = \{(x,y) | \, x \in A \wedge y \in A \wedge (x - y \geq 1)\}$,

$R_3 = \{(x,y) | \, x \in A \wedge y \in A \wedge \neg(x - y = 0)\}$,

$R_4 = \{(x,y) | \, x \in A \wedge y \in A \wedge \forall u \, \exists v \, (x + u = y + v)\}$ where u and v are integers, and

$R_5 = \{(x,y) | \, x \in A \wedge y \in A \wedge \exists u \, \forall v \, (x + u = y + v)\}$ where u and v are integers.

(i) List all the elements in R_1 and R_2.

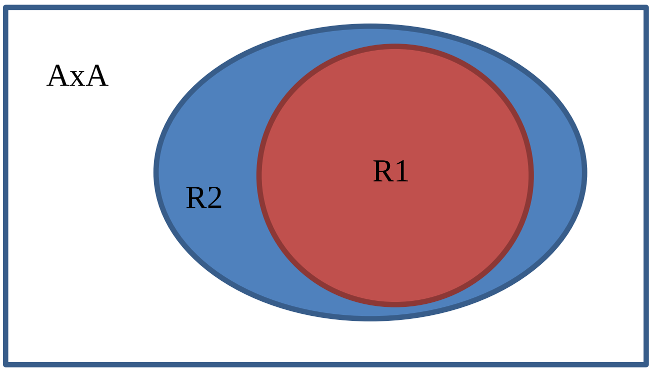
[4 marks]

(C): $R_1 = \{(2,1), (3,2)\}, R_2 = \{(2,1), (3,1), (3,2)\}$. 2 marks for each relation.

(ii) Draw a Venn diagram to show the relations among R_1, R_2 and $A \times A$.

[2 marks]

2 marks for the Venn diagram.



(iii) Represent relations R_3 using Boolean matrices. Explain whether they are symmetric.

[4 marks]

(C): $R3 = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$, they are both symmetric. 2 marks each.

(iv) Are sets R_4 and R_5 finite? Is it true that $R_4 = R_5$? Show your work.

[6 marks]

(C+P): both R4 and R5 are finite as they are the subsets of AxA. 1 marks each. However, R4 is not equal to R5 (1 mark). For R4, since there exists v such that v = x+u-y for any x,u,y, therefore R4=AxA (1.5 marks). For R5, u=y+v-x. for fixed x,y, u varies with v. So there does not exist a u that remains the same when v varies. Therefore, R5={}. (1.5 marks)

(b) Let f and g be functions from $A = \{1, 2, 3, 4\}$ to $B = \{a, b, c, d\}$ and from B to A, respectively, with $f(1) = d$, $f(2) = c$, $f(3) = a$, and $f(4) = b$, and $g(a) = 2$, $g(b) = 1$, $g(c) = 3$, and $g(d) = 2$.

(i) Is f one-to-one? Is g one-to-one?

[4 marks]

(k+C): since every element in B has a different pre-image in A, thus f is one-to-one. However, for g, 2 in A has two pre-images in B, i.e., a and d. Thus g is not one-to-one. 1 mark for final answer, 1 mark for the explanation in each case.

(ii) Is f onto? Is g onto?

[4 marks]

(K+C): for f, since every element in B has at least one pre-image in A, thus f is onto. For g, 4 in A does not have a pre-image in B, therefore it is not onto. 1 mark for final answer and 1 mark for the explanation in each case.

(iii) Does either f or g have an inverse? If so, find this inverse.

[4 marks]

(K+C): since f is both one-to-one and onto, then f is bijective and thus have an inverse. (1 mark). It is f-1(d)=1, f-1(c)=2, f-1(a)=3, f-1(b)=4. (1 mark). Since g is not onto, it not bijective either, thus no inverse. (1 mark for final anser, 1 mark for explanation)

(iv) Let $S_f: P(A) \to P(B)$, where $S_f\,(X) \;=\; \{y|\forall x \in X, y = f(x)\}$. $P(A)$ and $P(B)$ are power sets.

Determine the values of $S_f(\{1,2\})$ and $S_f(\emptyset)$.

[5 marks]

(C): Sf({1,2})=f{1,2}={d,c} (2 marks). Sf({})={} since no elements in null set thus no elements being mapped to B. (2 marks). 1 mark for correct formal representation.

2. Questions on Numbers, Sequences, Matrices, and Induction.

(a) Assume a, b, m, n are integers, and $a \equiv b (\mathrm{mod}\ m)$ (i.e., a is congruent to b modulo m).

(i) Show that if $n \mid m$, then $a \equiv b (\mathrm{mod}\ n)$.

[6 marks]

(C): m = kn (1.5 marks), from (i) we know a-b = i*m = i*k*n = (ik)n (3 marks). Thus a is congruent to b modulo n (1.5 marks).

(ii) Let $a = 3^7 \cdot 5^3 \cdot 7^3$, $b = 2^{11} \cdot 3^5 \cdot 5^9$. What are the greatest common divisor and the least common multiple of a and b respectively? Show your work.

[6 marks]

(K): gcd(a,b) = 2^0 * 3^5 * 5^3 * 7^0 (1.5 mark for correct final answer, 1.5 mark for intermediate representation). Lcm(a,b) = 2^11 * 3^7 * 5^9 * 7^3 (1.5 mark for correct final answer, 1.5 mark for intermediate representation).

(b) List the first 4 terms of each of the following sequences.

(i) The n-th term is given by $\lfloor n/2 \rfloor + \lceil n/2 \rceil$.

[6 marks]

(C): {1, 2, 3, 4}, 1.5 marks for each correct answer.

(ii) The sequence whose first two terms are 1 and 5 and each subsequent term is the sum of the two previous terms.

[6 marks]

(K+C): {1,5,6,11} 1.5 marks for each correct answer.

(c) Suppose that $A = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$, where a and b are real numbers. Show that $A^n = \begin{bmatrix} a^n & 0 \\ 0 & b^n \end{bmatrix}$ for every positive integer n, using mathematical induction.

[9 marks]

(K+C): for n=1, A^1 = [a^1, 0; 0, b^1]; (2 mark)
Assume n=k, that A^k = [a^k, 0; 0, b^k] (2 mark)
We have for n=k+1, A^(k+1)=A^k * A (2 mark)
= [a^k, 0; 0, b^k] * [a^1, 0; 0, b^1] (1 mark)
= [a^(k+1), 0; 0, b^(k+1)] (1 mark)
Thus for all positive n, we have A^n = [a^n, 0; 0, b^n] (1 mark)

3. Questions on Sets, Functions, Counting, and Probability.

(a) Let the set $A = \{1, 2, \ldots, 5\}$ and the set $B = \{0, 1\}$. Let $f: A \to B$ denote a total function from A to B, i.e., all elements in A is mapped to an element in B.

(i) How many possible functions f are there from A to B?

[6 marks]

(K+C): each element in A can be assigned either 0 or 1 in B, i.e. two possibilities. (3 marks). Thus the total number of different functions of f is 2^5 = 32. (3 marks)

(ii) Among the results in (i), how many functions are onto?

[6 marks]

(C): the number of onto functions are the result of (i) minus the number of functions that are not onto. There are only two cases that are not onto, i.e., all elements in A are mapped to 0, or all elements in A are mapped to 1. Thus the answer is 32-2 = 30. Any alternative solutions are also acceptable. 3 marks for final correct answer, 3 marks for explanation.

(iii) Among the results in (i), how many functions are there that assign 0 to both 1 and 5?

[6 marks]

(C): 1 and 5 have already been assigned values. Only 2, 3, 4 are free variables to be assigned values from {0,1}. So the answer is 2^3 = 8 (i.e., 2 possibilities for each of the three free variables). 3 marks for final correct answer, 3 marks for general explanation.

(b) A pair of dice is rolled. The probability that a 4 appears on the first dice is $2/7$, and the probability that a 3 appears on the second dice is $2/7$. Each of the other outcomes has probability $1/7$. What is the probability of 7 appearing as the sum of the numbers on the two dice?

[8 marks]

(C+P): There are in total 6 possibilities to have a sum of 7 from two rolled dies, i.e., (1,6), (2,5), (3,4), (4,3), (5,2), and (6,1). (3 marks). Except the case (4,3) with the probability as 2/7 * 2/7 = 4/49 (1 marks), the rest 5 cases are 1/7 * 1/7 = 1/49 (3 marks). The total probability is then 4/49 + 5* 1/49 = 9/49. (2 marks)

(c) Show that if seven integers are selected from $\{1,2,3,4,5,6,7,8,9,10\}$, there must be at least two pairs of these integers with the sum 11.

[8 marks]

(C): There are 5 cases that a pair of integers within 1 to 10 will add up to 11, i.e., (1,10), (2,9), (3,8), (4, 7), (5, 6) (3 marks). The worst scenario of selecting 7 numbers will include one number from each of the five pairs (3 marks), and the other two numbers will have to fall into either two of the pairs (2 marks). Thus there will be at least two pairs with sum 11.